The right sex for the mechanism: spinal microglial maintenance in hard flaccid syndrome

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Thirteenth in a series; see companion papers on the neural, effector, sensory and structural conjectures.

Abstract

Background. Central sensitization is described as reversible in principle, yet hard flaccid syndrome (HFS) rarely remits. The conjecture examined here proposes that spinal microglial activation following the initiating injury converts a transient sensitized state into a persistent one — a maintenance mechanism rather than an initiating one. Its author calls it a placeholder for a mechanism rather than a mechanism, and rates it among the weakest in the set.

One finding should raise that rating substantially, and it is not cited. Spinal microglial signalling in persistent pain is *sexually dimorphic*. Microglia are required for mechanical pain hypersensitivity in male mice but not in female mice, which use adaptive immune cells instead; microglial inhibition is analgesic in males and not females; and the effect is testosterone-dependent, being absent in castrated males and in prepubertal mice of both sexes and reinstated by testosterone. Hard flaccid syndrome is, so far as the literature records, an exclusively male condition of young men. **Of all the neuroimmune mechanisms available, the microglial one is the one that fits this population.**

Three problems remain. The conjecture's low specificity is real: microglial activation accompanies persistent pain of almost any origin and predicts little unique to HFS. Its mechanism is stated for the dorsal horn, whereas the defining sign of HFS is autonomic — whether microglial activation reaches the sympathetic preganglionic population is not addressed and appears not to have been studied. And its proposed test may be the wrong instrument: translocator protein positron emission tomography is not microglia-specific, requires genotyping that excludes 5–10% of people, has produced conflicting and very small human spinal-cord datasets, and has been reported in animals to detect acute but not diffuse chronic microglial activation — which is precisely the state this conjecture proposes.

An argument worth making carefully. Human trials of microglial inhibitors have largely disappointed, but they have generally not been stratified by sex. If the microglial contribution is male-specific, mixed-sex trials would be diluted for the male effect — and an all-male condition is an unusually favourable population in which to look. We set this out as a design observation, not as a treatment rationale, and note the reasons for caution.

Keywords: hard flaccid syndrome; microglia; neuroinflammation; sexual dimorphism; testosterone; TSPO PET; minocycline.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. **Nothing in this paper is a rationale for taking minocycline or any other antibiotic, anti-inflammatory or immunomodulatory drug.** Minocycline carries risks including drug-induced lupus, hepatotoxicity, intracranial hypertension and pigmentation changes, and antibiotic use without indication contributes to resistance. Nothing here should influence clinical management.

1. Introduction

Central sensitization is characterised as a prolonged but *reversible* increase in the excitability and synaptic efficacy of neurons in central nociceptive pathways **[A]** [8]. Hard flaccid syndrome (HFS) is not reversible in most patients: symptom resolution was documented in only 2 of the 10 patients whose outcomes were reported in the 2025 systematic review **[B]** [1]. Something has to close that gap.

Two companion papers propose candidates: a persisting peripheral generator feeding the central state [6], and the central state itself [5]. The conjecture examined here proposes a third — that spinal microglial activation following peripheral nerve injury produces enduring changes in dorsal horn neurons, converting a transient sensitization into a self-maintaining one [4, 9].

Its author is unusually frank about its weakness: neuroinflammation is invoked so widely in chronic pain that its explanatory value is diluted, it predicts little specific to HFS, and it should be regarded as a placeholder for a mechanism rather than a mechanism [4].

That assessment is largely right, and we do not overturn it. But it omits a finding that bears on this condition more directly than on almost any other to which the microglial hypothesis has been applied, and that changes what kind of study would be worth doing.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *spinal microglia persistent pain*, *sex differences microglia pain*, *TSPO PET chronic pain spinal cord*, *minocycline neuropathic pain trial*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic. We searched specifically for the current status of translocator protein imaging as an instrument and for sex-dependence of microglial pain signalling, and Sections 4 and 5.3 report what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence, or a principle established across independent laboratories and preparations.
[B]	Single human study, small human imaging series, or human data with substantial methodological limitation.
[C]	Experimental animal evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. Background: microglia as a maintenance mechanism

Peripheral nerve injury activates spinal microglia, which release mediators including brain-derived neurotrophic factor and proinflammatory cytokines, producing enduring changes in dorsal horn neurons that contribute to central sensitization and persistent pain **[C]** [9, 10]. P2X4 receptors induced in spinal microglia gate tactile allodynia after nerve injury **[C]** [VERIFY] [11]; microglial heterogeneity in neuropathic pain has since been characterised in detail **[C]** [VERIFY] [12].

The role this fills is exactly the one HFS requires. It is a *maintenance* mechanism, not an initiating one — and the puzzle in this condition is not why symptoms start but why they never stop [4].

4. The finding that should raise this conjecture's rating

Spinal microglial signalling in persistent pain is sexually dimorphic, and the dimorphism runs in the direction that favours this hypothesis.

Microglia are not required for mechanical pain hypersensitivity in female mice; female mice achieve similar levels of hypersensitivity using adaptive immune cells, likely T lymphocytes **[C]** [13]. Spinal cord Toll-like receptor 4 mediates inflammatory and neuropathic hypersensitivity in male but not female mice **[C]** [VERIFY] [14]. Deletion of microglial brain-derived neurotrophic factor prevented full allodynia in male but not female mice **[C]** [13].

The hormonal dependence is the part that matters most here. Minocycline — used in that work as a microglial inhibitor — was **ineffective in castrated male mice and in young (4-week-old) mice of both sexes, and its efficacy was reinstated by testosterone** **[C]** [13]. The male pathway is androgen-dependent, not merely male-associated. The same male-specific pattern has since been reported for minocycline analgesia in another model **[C]** [VERIFY] [16], and regulatory T cells have been shown to suppress a microglial pain pathway in females that is otherwise competent **[C]** [VERIFY] [15].

4.1 Why this matters for hard flaccid syndrome specifically

Hard flaccid syndrome, as described in the published literature, occurs in men **[B]** [1, 2]. The reported population is young — mean age 27.4 years, range 16–42 across studies [2, 1] — and therefore, in general, post-pubertal with normal androgen status [DATA NEEDED: TESTOSTERONE HAS NOT BEEN SYSTEMATICALLY REPORTED IN HFS COHORTS].

Three consequences follow **[D]**:

1. **The mechanism matches the population.** Of the neuroimmune mechanisms available, the microglial one is the one the sex-difference literature says operates in males. Applying a microglial hypothesis to an all-male condition is better targeted than applying it to a mixed-sex chronic pain population.
2. **The usual generalisability criticism is inverted.** Preclinical pain research has historically used male rodents, which is normally a limitation when extrapolating to mixed-sex human populations. Here the animal literature and the patient population are matched on the variable that turns out to determine the mechanism.
3. **It predicts an androgen interaction.** If the pathway is testosterone-dependent, HFS phenotype or severity might relate to androgen status. We have not seen this examined, and note that hormonal studies in HFS are described as normal [1] without, so far as we can establish, systematic reporting of values [DATA NEEDED].

We are careful about what this does and does not establish. It does not make the conjecture more likely to be *true* in HFS; no microglial measurement has been made in this condition, and species extrapolation from mouse pain behaviour to human genital symptoms is a long reach **[D]**. What it does is make the conjecture better *targeted* — and it removes one of the standard objections to applying rodent neuroimmune findings to a clinical population.

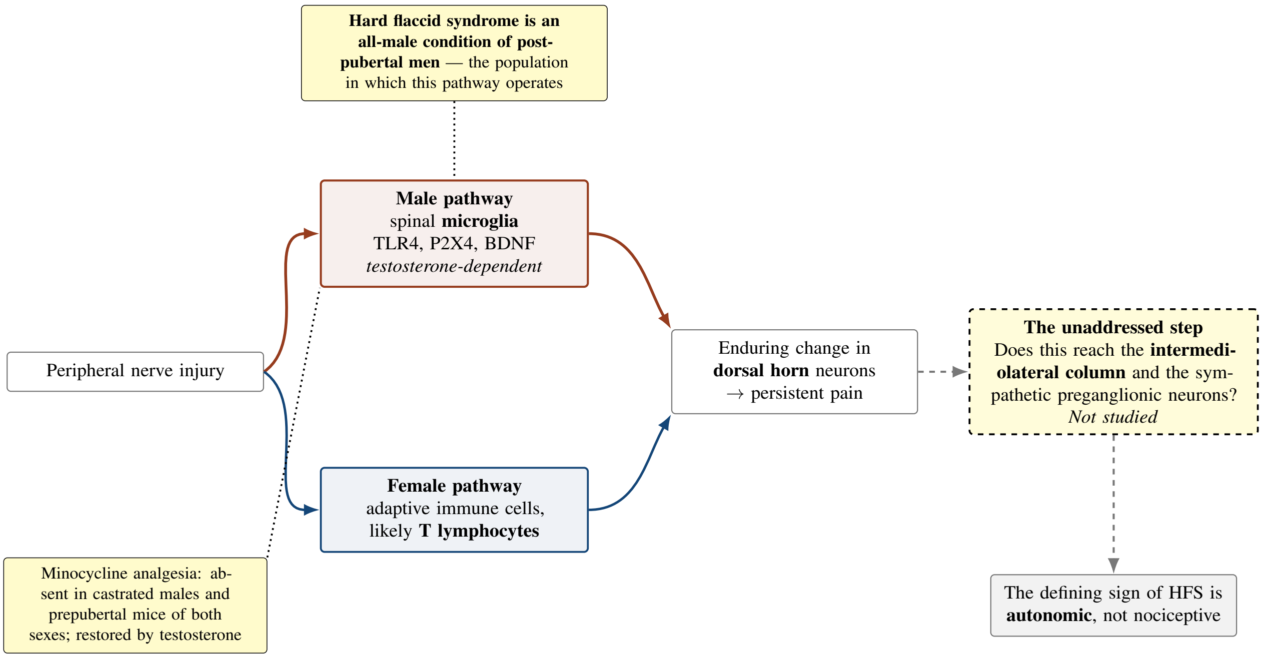

Figure 1. The sexually dimorphic neuroimmune pathway, and the step the conjecture does not address. Microglia mediate mechanical pain hypersensitivity in male mice; female mice use adaptive immune cells instead, and microglial inhibition is analgesic only in males and only in the presence of testosterone [13, 14]. Hard flaccid syndrome is an all-male condition of post-pubertal men, which makes the microglial hypothesis better targeted here than in a mixed-sex pain population. The dashed portion is the gap: the mechanism is characterised for the dorsal horn, whereas the defining sign of this condition is autonomic, and whether microglial activation modifies drive to sympathetic preganglionic neurons appears not to have been studied [5].

5. Three problems that remain

5.1 The specificity problem is real

Elevated translocator protein signal has been reported in chronic low back pain, fibromyalgia, migraine and other persistent pain conditions **[B]** [VERIFY] [17]. A mechanism that accompanies persistent pain of almost any origin does not, by its presence, explain any particular one. The conjecture's self-assessment stands: as a claim about HFS, it predicts almost nothing that would distinguish HFS from any other chronic pain state **[D]**.

There is a partial mitigation. The spatial distribution of the signal has been reported to exhibit a degree of disorder specificity **[B]** [VERIFY] [17], and thalamic neuroinflammation has been proposed as a reproducible and discriminating signature for chronic low back pain **[B]** [VERIFY] [18]. If distribution carries information, a distribution study in HFS would be more informative than a presence/absence study — but that raises the sample size and the cost.

5.2 The autonomic gap

The mechanism as stated concerns the dorsal horn and persistent pain [9, 4]. Hard flaccid syndrome is named for an autonomic sign. For this conjecture to explain the condition rather than one of its symptoms, microglial activation would have to modify the excitability of, or the drive to, sympathetic preganglionic neurons in the intermediolateral column — the same transfer step that the central maintenance conjecture requires and cannot supply [5].

We identified no study examining microglial modulation of sympathetic preganglionic neurons after peripheral nerve injury [DATA NEEDED]. This is a substantial gap, it is shared with the central sensitization hypothesis, and it means that **as stated, this conjecture explains the pain of HFS and not the hard flaccid state**.

5.3 The instrument problem

The conjecture proposes translocator protein positron emission tomography as the obvious approach, noting that it is expensive and available in few centres [4]. The difficulties are more fundamental than cost.

1. **It is not microglia-specific.** Translocator protein is expressed by both microglia and astrocytes, precluding cell-type attribution in current studies **[B]** [VERIFY] [22]. A positive result would show glial activation, not microglial activation.
2. **It requires genotyping and excludes some people entirely.** The rs6971 polymorphism affects ligand binding affinity, necessitating genotyping and potentially excluding approximately 5–10% of individuals as low-affinity binders **[B]** [VERIFY] [22]. Second-generation ligands are sensitive to this polymorphism; third-generation ligands aim to reduce that sensitivity, and there is still no ideal ligand **[B]** [VERIFY] [21, 23].
3. **The human spinal-cord data are sparse and conflicting.** In one comparison of neuropathic pain patients and controls ($n = 5$ per group), translocator protein imaging revealed no observable spinal neuroinflammation; in another, patients with chronic radicular pain ($n = 19$ versus 10 controls) showed increased tracer uptake in the ipsilateral neuroforamina and spinal cord segments **[B]** [VERIFY] [19, 20].
4. **It may detect the wrong state.** In rats, translocator protein imaging detected acute neuroinflammation but not diffuse chronically activated microglia **[C]** [VERIFY] [24]. **Chronic, diffuse microglial activation is precisely what this conjecture proposes**, and if that finding generalises, the proposed instrument would be blind to the proposed lesion.

6. The hypothesis

Formal statement

H13 (neuroinflammatory maintenance). Peripheral nerve injury in hard flaccid syndrome activates spinal glia, principally microglia, whose signalling to neurons converts a transient sensitized state into a self-maintaining one, accounting for the chronicity of the condition.

H13-nociceptive (what is currently supported). The microglial state maintains the sensory symptoms — pain, dysesthesia, paresthesia. Prediction: glial activation at the relevant lumbosacral segments; sensory symptoms respond to microglial modulation.

H13-autonomic (what the condition requires, and what is unstudied). The microglial state also maintains elevated drive to sympathetic preganglionic neurons. Prediction: the hard flaccid state itself responds to microglial modulation. **No evidence exists for the transfer step this requires.**

H13-null. Glial activation in HFS is absent, or is present at the level found in any chronic pain condition and carries no explanatory weight.

Auxiliary assumptions:

- **A1 (persistence).** Microglial activation persists for years. Activation in animal models is characteristically time-limited, and the durability of a glial state over the timescale of this condition is not established [DATA NEEDED].
- **A2 (the transfer step).** Microglial signalling reaches the sympathetic preganglionic population. Not studied in any species.
- **A3 (androgen status).** The population has the hormonal profile in which this pathway operates. Plausible on age

grounds but not documented [DATA NEEDED].

7. Predictions and tests

7.1 If imaging is attempted, image the right level with the right design

Prediction P1. Men with HFS will show elevated translocator protein signal at lumbosacral spinal cord segments and neuroforamina relative to matched controls.

The lumbosacral level is the right target for a specific reason: 76% (16/21) of men with HFS who underwent lumbar magnetic resonance imaging in one congress abstract had a lumbosacral annular tear **[B]** [3, 1], and the one positive human spinal-cord finding in this literature is from chronic *radicular* pain, where uptake was localised to ipsilateral neuroforamina and spinal cord segments [VERIFY] [20]. **The protocol therefore already exists for an anatomically analogous population**, which is the strongest practical argument for attempting it.

Requirements: rs6971 genotyping with pre-specified handling of low-affinity binders; a third-generation ligand if available; matched controls scanned identically; and pre-specification of whether the outcome is presence, magnitude or spatial distribution. Given the sample sizes in the existing spinal literature (5 and 19 patients), **a study powered to detect a modest difference would be larger than anything yet attempted in this field** [DATA NEEDED].

7.2 The argument about trial design, stated carefully

Human trials of microglial inhibitors — minocycline in particular — have largely disappointed in neuropathic pain **[B]** [VERIFY: WE DID NOT RETRIEVE A SYSTEMATIC REVIEW OF THESE TRIALS AND THIS CHARACTERISATION SHOULD BE CHECKED] [19]. That record is usually read as evidence against the microglial hypothesis in humans.

There is an alternative reading worth recording. If the microglial contribution to persistent pain is male-specific and androgen-dependent [13], then **a mixed-sex trial of a microglial inhibitor would be diluted for the male effect**, and could return null even if a real male-specific effect existed **[D]**. Trials in this field have generally not been designed or analysed with that in mind.

Hard flaccid syndrome is an all-male condition of post-pubertal men. On this reasoning it would be an unusually favourable population in which to look for a microglial effect.

We set this out as an observation about trial design and stop there, for four reasons. Minocycline is a poor and non-selective microglial inhibitor with substantial off-target effects, so a positive result would not localise to microglia. Its safety profile is not trivial, including drug-induced lupus, hepatotoxicity and intracranial hypertension, and antibiotic exposure without indication has population-level costs. Translation from rodent pain behaviour to human genital symptoms has failed repeatedly across this series. And the transfer step to the autonomic limb — the thing that would make the hard flaccid state itself respond — has no evidence at all. **This is a reason to design future trials with sex in mind, not a reason for anyone to take an antibiotic.**

7.3 The cheaper things that should come first

Prediction P2. Quantitative sensory testing will show a central sensitization phenotype in men with HFS [5]. If it does not, H13 loses most of its motivation, since the glial state is proposed to maintain a sensitized state that would then not exist.

Prediction P3. Serum or cerebrospinal fluid inflammatory markers will differ from matched controls. We note this is a weak test: peripheral markers correlate poorly with central glial state, and a negative result would not exclude the hypothesis [DATA NEEDED].

Table 2. Predictions of H13 compared with the principal alternatives.

Observation	H13 (glial)	Central sensitization	Peripheral generator	Discriminating value
Elevated lumbosacral spinal TSPO signal	**Predicted**	Not predicted	Not predicted	**High, if the instrument works**
Central sensitization phenotype on QST	Predicted (required)	**Predicted**	Predicted	**None**
Sensory symptoms respond to microglial inhibition	**Predicted**	Not predicted	Not predicted	Moderate; agent non-selective
Hard flaccid state responds to microglial inhibition	Predicted under H13-autonomic only	Not predicted	Not predicted	**High, and unsupported by any mechanism**
Signal distribution differs from other chronic pain conditions	Predicted (partial mitigation of low specificity)	Not specified	Not specified	Moderate
Symptom severity relates to androgen status	**Predicted**	Not predicted	Not predicted	Moderate, and never examined

8. Alternative explanations

1. **A peripheral generator supplies the persistence instead.** Sodium channel remodelling produces continuing afferent input that maintains a central state without any glial contribution [6]. This is the direct competitor for the maintenance role, and it is cheaper to test.
2. **Glial activation is a marker, not a maintainer.** Elevated signal in chronic pain conditions is consistent with glial change being a consequence of persistent afferent traffic rather than its cause. No imaging study can distinguish these.
3. **The chronicity may not need explaining.** The low remission rate derives from case material and self-selected surveys, and men whose symptoms resolve leave patient communities and stop seeking care [7]. Ascertainment bias is a simpler account of apparent irreversibility than a self-maintaining glial state.
4. **Low specificity may be the whole story.** If glial activation accompanies persistent pain generally, finding it in HFS would place HFS among chronic pain conditions without explaining what makes it this one.
5. **The premise may be false.** No central sensitization phenotype has been demonstrated in HFS [5], and a mechanism proposed to maintain a state should not be investigated before the state itself.

9. Implications

No clinical implications are drawn, and we repeat that nothing here is a reason to take minocycline or any other immunomodulatory drug. Three research implications follow.

First, establish the state before investigating its maintenance. Quantitative sensory testing is non-invasive, uses a validated protocol and costs a fraction of a positron emission tomography study. If there is no sensitization phenotype, this conjecture has nothing to maintain.

Second, if imaging is attempted, target lumbosacral segments and neuroforamina, using the protocol developed for chronic radicular pain, with rs6971 genotyping and a realistic sample size.

Third, record sex-relevant variables in any future neuroimmune trial. The observation that microglial pain signalling is male-specific and androgen-dependent has implications for trial design across the pain field, and an all-male condition is an informative setting in which to act on it.

10. Limitations

1. **No evidence in HFS.** No glial measurement of any kind has been made in this condition.
2. **Low specificity is intrinsic**, not merely a matter of how the hypothesis is stated. Glial activation accompanies persistent pain broadly.
3. **The autonomic transfer step has no evidence in any species**, which means the conjecture as stated addresses the pain of HFS and not its defining sign.
4. **The proposed instrument may be blind to the proposed lesion**, given the report that translocator protein imaging detects acute but not diffuse chronic microglial activation.
5. **The sex-specificity argument is rodent behavioural data**, extrapolated to a human genital condition across a very large gap. It improves targeting, not plausibility.
6. **Androgen status in HFS is undocumented**, so even the population-matching argument rests on an assumption about age.
7. **The trial-design argument could be misread as a treatment rationale.** We have tried to close that off explicitly, but the risk is real in a patient community with few options.
8. **Persistence over years is not established** for a glial state in any model.
9. **Many citations are flagged unverified**, including the characterisation of the human minocycline trial record, which we could not confirm against a systematic review.
10. **Confirmation risk.** This paper argues that a conjecture its author rates low deserves a higher rating, on the basis of one finding. Readers should note that better targeting is not evidence, and that we have not moved the hypothesis any closer to being supported in this condition.

11. Refutation criteria

- **Substantially refuted.** Translocator protein imaging at lumbosacral segments, adequately powered and genotype-controlled, shows no elevation relative to matched controls.
- **Motivation removed.** Quantitative sensory testing shows no central sensitization phenotype in men with HFS.
- **Demoted to H13-nociceptive.** Microglial modulation improves sensory symptoms while leaving the hard flaccid state unchanged.
- **Weakened.** Elevated signal is found but is indistinguishable in magnitude and distribution from that in other chronic pain conditions.
- **Weakened.** Symptom severity shows no relationship to androgen status in a condition whose proposed mechanism is androgen-dependent.
- **Rendered unnecessary.** A peripheral generator is demonstrated and accounts for the persistence without a glial contribution [6].

12. Conclusion

This is the conjecture its author describes as a placeholder, and in most respects that description holds. Glial activation accompanies persistent pain of almost every origin; a mechanism that general does not explain a particular condition. The proposed mechanism is stated for the dorsal horn while the condition is named for an autonomic sign, and the step between them — microglial modulation of sympathetic preganglionic drive — appears never to have been studied. And the instrument proposed to test it is not microglia-specific, requires genotyping that excludes some people entirely, has produced conflicting results in spinal cord studies of five and nineteen patients, and has been reported in animals to detect acute but not diffuse chronic activation, which is exactly the state at issue.

One thing pulls the other way, and it is not small. Spinal microglial signalling in persistent pain is male-specific and testosterone-dependent: microglia are required in male mice and not female, microglial inhibition is analgesic only in males, and that analgesia disappears after castration and before puberty and returns with testosterone. Hard flaccid syndrome is an all-male condition of post-pubertal men. The usual complaint about extrapolating from male rodents to patients does not apply, because here the population and the preclinical literature are matched on the variable that determines which mechanism operates. Among the fourteen conjectures, this is the only one whose supporting animal literature is aligned with the patient population on a variable known to be mechanistically decisive.

That is an argument about targeting, not about truth, and we want to be clear that it moves the hypothesis no closer to being supported in this condition. What it does suggest is an order of work. Establish that a sensitized state exists, with a sensory testing protocol that costs a fraction of a scan. If it does, and if imaging is attempted, aim it at the lumbosacral segments where a radicular protocol already exists and where three quarters of one small HFS cohort had annular tears. And whenever the pain field next tests a microglial intervention in humans, design it with sex in mind — because if the rodent work is right, the trials that have already failed may have been asking a male question of a mixed-sex population.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** The characterisation of the human minocycline trial record in Section 8.2 was not confirmed against a systematic review and must be verified or removed, since the argument built on it is the most likely part of this paper to be misused.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [4]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. **Nothing in it supports the use of minocycline, other tetracyclines, or any anti-inflammatory or immunomodulatory agent for hard flaccid syndrome.** Minocycline carries risks including drug-induced lupus, hepatotoxicity, idiopathic intracranial hypertension, vestibular effects and permanent pigmentation, and antibiotic use without a clinical indication contributes to antimicrobial resistance. Positron emission tomography involves ionising radiation and is a research procedure requiring ethical approval.

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